WEB PROGRAMMING CHECKPOINT 1

PROJECT NAME: ONLINE CLEAN BOOK

I. PROJECT INFORMATION

1. Project Name

Clean Book is an e-commerce website project that focuses on selling books online. The website aims to provide customers with a convenient and easy-to-use platform where they can browse, search for, and purchase books from different categories.

The project focuses on making the online book-buying process simpler by providing clear product information, reasonable navigation, and an organized shopping experience.

2. Product Category

Product Category: Books - Online Bookstore

Clean Book specializes in selling books from different fields and categories, including:

No.	Book Category	Description
1	Literature and Novels	Fiction, novels, short stories, and literary works.
2	Self-help and Personal Development	Books about life skills, personal growth, and communication.
3	Economics and Business	Books about economics, business, management, and finance.

4	Education and Reference	Textbooks, study materials, and reference books.
5	Science and Information Technology	Books about science, technology, and computer science.
6	Children's Books	Picture books, educational books, and stories for children.

Products will be organized into categories to help customers find books more easily. Each book will have basic information such as its title, author, price, description, and available stock.

3. Target Customers

Clean Book targets the following customer groups:

Customer Group	Needs and Interests
Students	Looking for textbooks, reference materials, and books to support their studies.
Working Professionals	Interested in business, professional skills, and personal development books.
Book Lovers	Regularly searching for novels, literature, and books from different genres.

Target Age Group: Primarily 16-35 years old.

These customers are familiar with using the Internet to search for information, compare products, and shop online. They may prefer an online bookstore because it allows them to explore different books and make purchases without visiting a physical store.

4. Problem Statement

4.1. Existing Problems

Buying books can sometimes be inconvenient for customers, especially when they do not know exactly which book they want or where to find it.

Some common problems include:

- Customers may spend a lot of time searching for books that match their interests and needs.
- Finding books by title, author, or category may be difficult when product information is not well organized.
- Customers may not have enough information about a book's content, price, and availability before making a purchase.
- Visiting physical bookstores may be inconvenient for customers who live far away or have limited free time.

4.2. Proposed Solution

Clean Book aims to address these problems by developing an online bookstore with a simple and user-friendly interface.

The website will allow customers to:

- Browse books by category.
- Search for books by title or author.

- View detailed information about each book.
- Add books to a shopping cart and update quantities.
- Place orders online.
- View relevant product information before making a purchase.
- And more comfortable service if customers need to add in e-shop.

Through these features, Clean Book aims to reduce the time and effort required to buy books, make books easier to access, and provide customers with a more convenient online shopping experience.

II. TEAM MEMBERS AND RESPONSIBILITY ASSIGNMENT

1. Team Member List

The project team consists of four members. Each member is assigned specific tasks based on the project requirements and their responsibilities.

Member	**Role**	**Main Responsibility**
Lê Trọng Phát	Project Information	Prepare the project overview and identify the target customers and problems.
Đinh Trọng Phú	System Planning	Define the initial features, user flow, wireframe, technology stack, and development plan.
Trần Gia Triệu	Research and Analysis	Research similar e-commerce websites and analyze their features and user experience.

Trương Hoàng Đăng Khoa	Team Leader	Coordinate the team, manage the Git repository, review all work, and prepare the final report.

2. Responsibility Assignment

The tasks are divided among the four members as follows:

2.1. Lê Trọng Phát

Assigned Tasks:

- Define the project name.
- Identify the product category.
- Describe the target customers.
- Write the problem statement and proposed solution.

Expected Deliverables: Project information, product description, target customer groups, and problem statement.

2.2. Đinh Trọng Phú

Assigned Tasks:

- Prepare the initial feature list.
- Create a rough user flow or wireframe for the website.
- Propose the technology stack for the project.
- Prepare the development plan for the remaining weeks.

Expected Deliverables: Initial feature list, user flow or wireframe, technology stack, and development plan.

2.3. Trần Gia Triệu

Assigned Tasks:

- Analyze two or three similar e-commerce websites.
- Identify target customers and create one or more user personas.
- Compare important features of the reference websites.
- Select the features for Clean Book and explain the reasons for choosing them.
- Provide UI/UX observations and links to the reference websites.

Expected Deliverables: Competitor analysis, user personas, feature comparison, selected features with explanations, and UI/UX observations.

2.4. Trương Hoàng Đăng Khoa - Leader

Assigned Tasks:

- Prepare the team member list and responsibility assignment.
- Create and manage the team's Git repository.
- Set deadlines and monitor the progress of each member.
- Collect and review the work submitted by all members.
- Check the consistency and accuracy of the project content.
- Edit and improve sections when necessary, in coordination with the responsible members.

- Combine all sections into one complete report.
- Format and prepare the final PDF file.
- Check the report against the instructor's requirements before submission.

Expected Deliverables: Team member list, responsibility assignment, Git repository, reviewed project content, and final PDF report.

3. Team Collaboration

- All team members are responsible for completing their assigned tasks and submitting their work on time.
- The team leader will coordinate the overall progress and review the submitted content to ensure that all sections are consistent with the Clean Book Online project.
- If any section needs correction or improvement, the team leader will discuss it with the responsible member before making changes.
- All members will participate in the final review to ensure that the report is complete and ready for submission.

III. INITIAL SYSTEM DESIGN AND DEVELOPMENT PLAN

1. Initial Feature List

The Clean Book project aims to develop an online bookstore that provides customers with a convenient platform to search for, view, and purchase books. The website initially focuses on three main product categories: textbooks, novels, and self-help books.

The system is divided into two main user groups: Customers and Administrators.

1.1. Customer Features

No.	Feature	Description
1	Home Page	Displays featured books, new arrivals, and book categories.
2	View Book Listings	Displays the books currently available on the website.
3	Search Books	Allows customers to search for books by title or author.
4	Filter by Category	Allows customers to filter books by category, such as textbooks, novels, and self-help books.
5	View Book Details	Displays book information, including images, price, author, publisher, and description.
6	Shopping Cart	Allows customers to add books, update quantities, and remove items from their cart.
7	Place Order	Allows customers to enter delivery information and confirm their orders.
8	Register / Login	Allows customers to create an account and log in to access personal features.
9	View Order History	Allows customers to review their previous orders and order information.

1.2. Administrator Features

No.	Feature	Description
1	Manage Books	Allows administrators to add, edit, delete, and view book information.
2	Manage Authors	Allows administrators to add, edit, and delete author information.
3	Manage Publishers	Allows administrators to add, edit, and delete publisher information.
4	Manage Categories	Allows administrators to create, update, and delete book categories.
5	Manage Orders	Allows administrators to view customer orders and update their statuses.

1.3. Priority Features

During the initial development stage, the team will prioritize the following core features:

- Display book listings and detailed book information.
- Search for books and filter them by category.
- Manage books, authors, and publishers.
- Add books to the shopping cart and update quantities.
- Place orders using basic delivery information.

- Manage customer orders.

Advanced features such as online payment, book reviews, and product recommendations may be considered for future development if time and resources permit.

2. User Flow

The Clean Book website is designed with a simple and user-friendly interface. Its main purpose is to help customers find suitable books and complete their purchases conveniently.

2.1. Customer User Flow

The customer shopping process consists of the following steps:

Step 1: The customer visits the Clean Book homepage.

Step 2: The customer searches for books or browses available book categories.

Step 3: The customer selects a book to view its detailed information.

Step 4: The customer adds the selected book to the shopping cart.

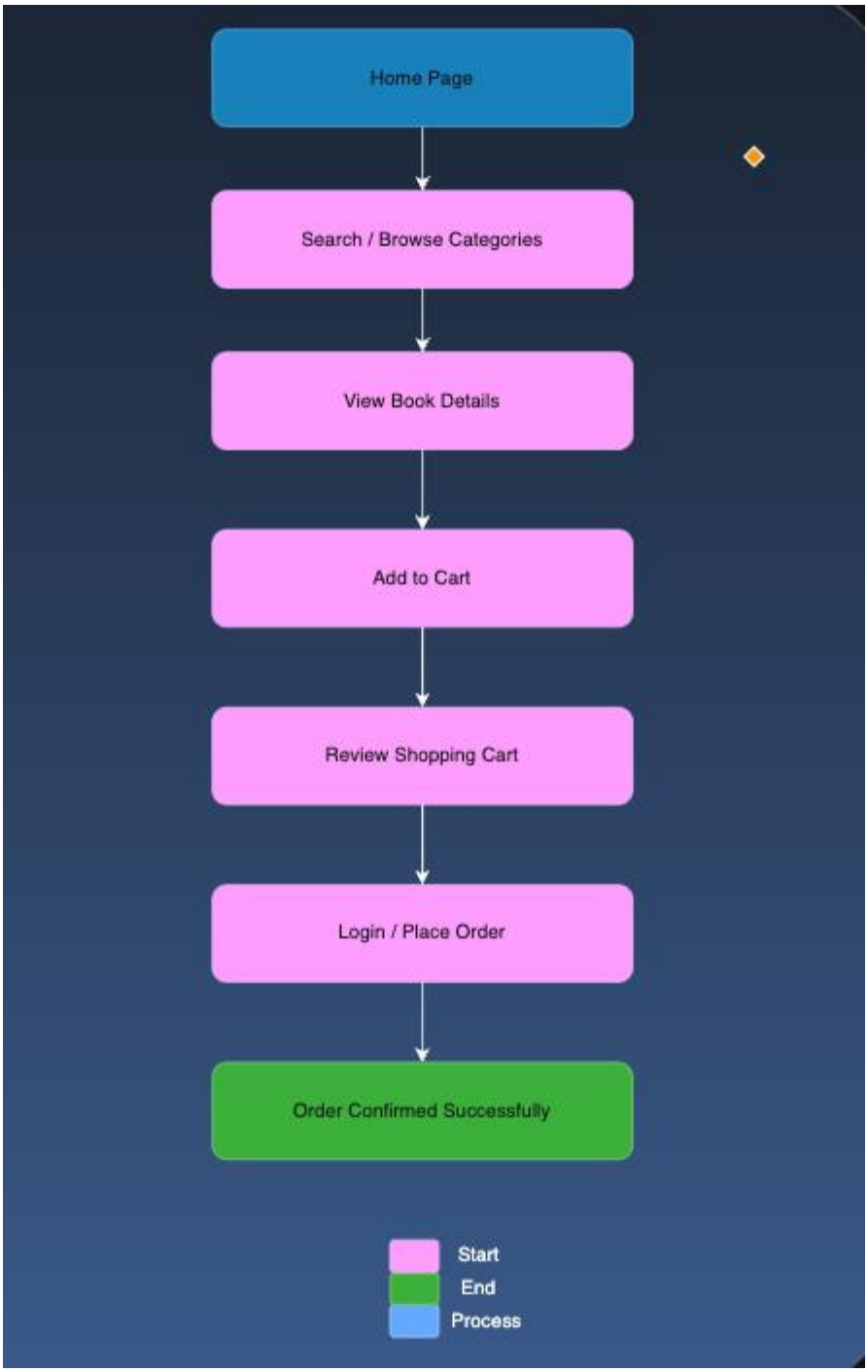
Step 5: The customer reviews the shopping cart and checks the selected books.

Step 6: The customer logs in or proceeds with the order process.

Step 7: The customer completes the ordering process.

Step 8: The system confirms that the order has been placed successfully.

Customer User Flow Diagram:



2.2. Administrator User Flow

The administrator uses the management interface to maintain website information and process customer orders.

Step 1: The administrator logs in to the system.

Step 2: The administrator accesses the Admin Dashboard.

Step 3: The administrator selects a management function, such as managing books, authors, publishers, categories, or orders.

Step 4: The administrator performs the required actions, such as adding, editing, deleting, or viewing records.

Step 5: The system saves the changes to the database and updates the information displayed on the website.

Administrator User Flow:

Admin Login → Admin Dashboard → Select Management Function → Add / Edit / Delete / View Data → Save Changes to Database → Display Updated Information.

2.3. Proposed Interface Design

The website is expected to include the following main pages:

Page	Description
Home Page	Displays the website logo, search bar, category menu, featured books, and new arrivals.
Book Listing Page	Displays available books with category filters and search options.
Book Detail Page	Displays the book image, price, author, publisher, description, and Add to Cart button.
Shopping Cart Page	Displays selected books, quantities, and the total price.
Checkout Page	Allows customers to enter delivery information and confirm their orders.
Customer Account Page	Allows customers to view their account information and order history.
Admin Dashboard	Provides access to book, author, publisher, category, and order management functions.

The interface will follow a modern and user-friendly design, with a clear layout, harmonious colors, and responsive support for both desktop computers and mobile devices.

3. Proposed Technology Stack

The team plans to use common web development technologies that are suitable for the project's scope and the members' current technical knowledge.

Technology	Purpose
HTML5	Builds the structure and content of web pages.
CSS3	Designs the website layout, colors, and visual appearance.
JavaScript	Handles user interactions and basic form validation.
Bootstrap 5	Supports responsive interface design.
PHP	Handles server-side logic, authentication, and website functions.
MySQL	Stores and manages books, authors, publishers, customers, and order data.
XAMPP	Provides a local development environment for running and testing the website.
Git & GitHub	Manage source code and support team collaboration.
Visual Studio Code	Provides an environment for writing and editing source code.

3.1. System Architecture

- The system is expected to follow a basic Client–Server Architecture, consisting of three main components:
- Frontend: Uses HTML, CSS, Bootstrap, and JavaScript to display the website interface and handle user interactions.
- Backend: Uses PHP to process user requests, implement business logic, and communicate with the database.

- Database: Uses MySQL to store and manage system information.

When a customer interacts with the website, the frontend sends requests to the PHP backend. The backend processes these requests and retrieves or updates the necessary information in the MySQL database before returning the results to the frontend.

3.2. Proposed Database Structure

The initial database is expected to include the following main tables:

Table	**Description**
Books	Stores book information, including book ID, title, price, description, and stock quantity.
Authors	Stores author information.
Publishers	Stores publisher information.
Categories	Stores book categories and their descriptions.
Users	Stores customer and administrator accounts, including login information and user roles.
Carts	Stores shopping cart information associated with customers.
Cart_Items	Stores the books and quantities added to each shopping cart.
Orders	Stores customer order information, delivery details, total amount, and order status.
Order_Details	Stores the books and quantities included in each order.

The tables will be connected using primary keys and foreign keys to maintain data relationships and consistency.

3.3. Reasons for Technology Selection

The team selected these technologies because they are accessible, have extensive learning resources, and are suitable for developing a small-scale e-commerce website.

HTML, CSS, and JavaScript support the development of an interactive and user-friendly interface. PHP and MySQL provide the necessary tools for server-side processing and data management. Bootstrap helps create a responsive design, while GitHub supports collaboration and source code management among team members.

4. Development Plan

The team plans to develop the project through several stages, starting with requirement analysis and interface design, followed by feature implementation, testing, and final improvements.

The following schedule covers five expected development weeks after Checkpoint 01. It may be adjusted according to the actual course schedule and project progress.

Timeline	Main Tasks	Expected Results
Week 1	Analyze requirements, design the website interface, create the initial database design, and set up the GitHub repository.	Complete the initial design and project structure.
Week 2	Develop the homepage, book listing page, book detail page, and responsive interface.	Complete the basic customer-facing interface.

Week 3	Develop the MySQL database and implement book, author, publisher, and category management.	Complete the main data management functions.
Week 4	Implement user registration and login, shopping cart, order placement, and order management.	Complete the basic online shopping functions.
Week 5	Integrate all components, test the system, fix errors, improve the interface, prepare the report, and organize the final demonstration.	Complete a basic working website that is ready for presentation.

4.1. Development Approach

The team will follow a phased development approach in which each member is responsible for assigned tasks.

The team will apply the following practices:

- Use GitHub to store source code and track changes.
- Agree on the project folder structure, naming conventions, and coding standards.
- Test each feature after its implementation.
- Communicate regularly to monitor progress and resolve technical issues.
- Integrate completed components and check their compatibility.
- Review the system before the final demonstration.

4.2. Expected Outcomes

After completing the project, the team expects to deliver a basic online bookstore with a user-friendly interface that allows customers to search for, view, and order books online.

The administration system will support the management of books, authors, publishers, categories, and customer orders.

Through this project, team members will also gain practical experience in frontend development, backend programming, database design, software testing, and teamwork using GitHub.

The final product will focus on implementing the essential functions of an online bookstore within the project's available time and resources.

5. Git Repository

The team created a GitHub repository named WEB PROGRAMMING to store project documents, manage source code, and track development progress.

The repository is used as a shared workspace where team members can upload their assigned work, update project files, and collaborate throughout the development process.

Git commits are used to record changes and track the team's progress from the initial planning stage.

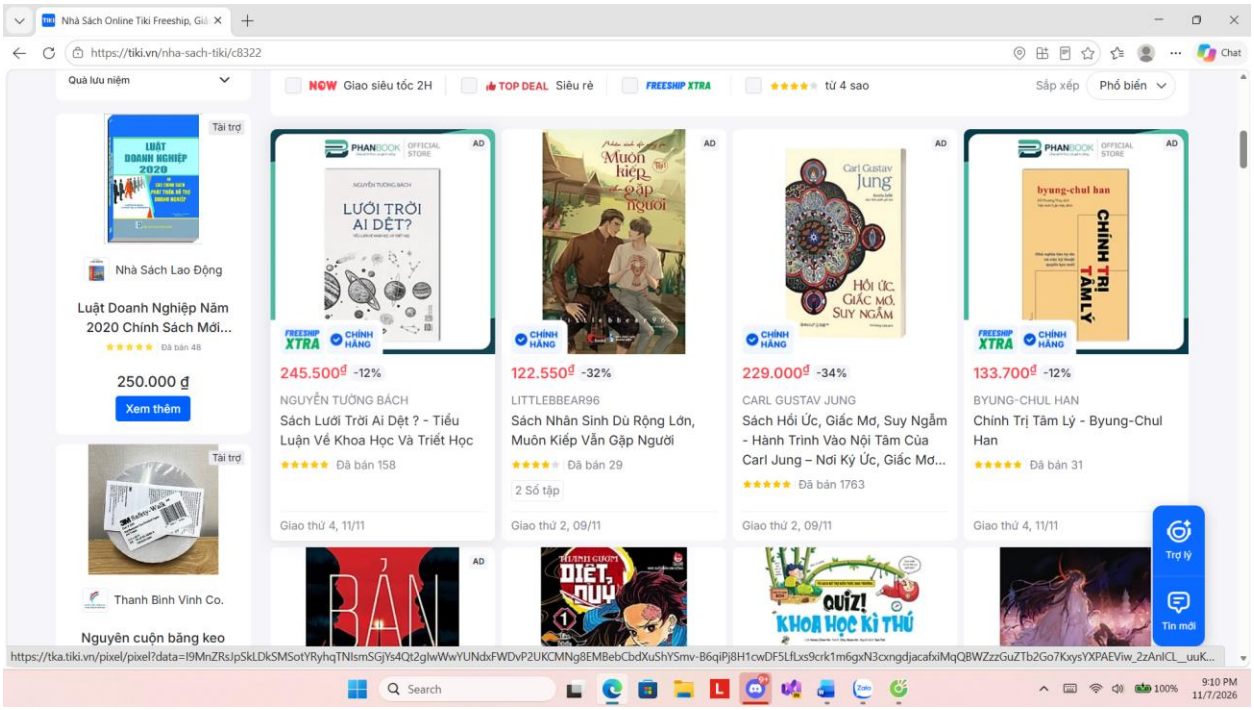
GitHub Repository: https://github.com/users/HanLapLHDC/projects/1

IV. REFERENCE WEBSITES AND FEATURES FROM FAMOUS ONLINE CLEAN BOOK

1. Analysis of Reference E-commerce Websites

1.1. Tiki

Website: https://tiki.vn/



Main Features:

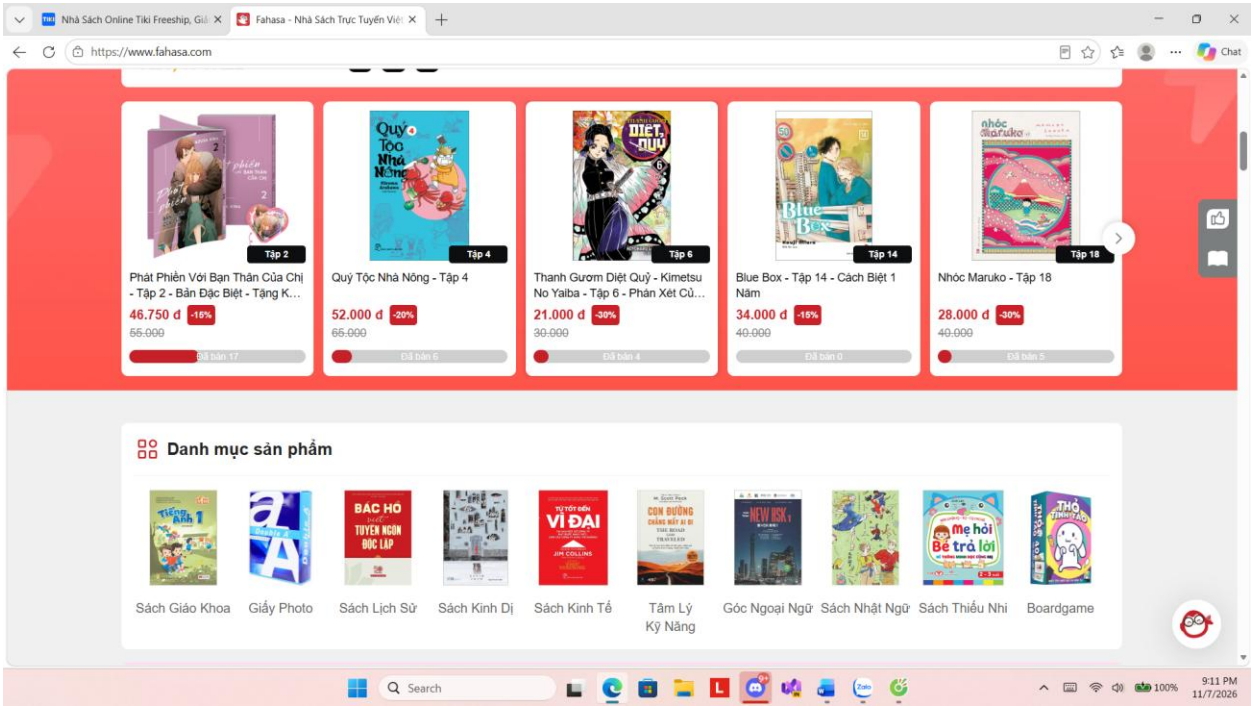
- Product Search: Allows customers to search for products by name or keyword.
- Product Categories: Organizes products into different categories.
- Product Details: Displays product images, prices, descriptions, and related information.
- Shopping Cart: Allows customers to save products before purchasing.
- Order Placement: Supports the online shopping process.

Key Characteristics:

Tiki is a multi-category e-commerce platform that serves different customer groups. Its wide range of products provides customers with various shopping options.

1.2. Fahasa

Website: https://www.fahasa.com/



Main Features:

- Book Search: Supports searching by book title or keyword.
- Product Categories: Organizes books and related products into different categories.
- Product Details: Displays book images, prices, descriptions, and other information.
- Shopping Cart: Allows customers to save products they want to purchase.

- Order Placement: Supports online book purchasing.

Key Characteristics:

Fahasa focuses on books, stationery, and related products. Its specialized product categories make it more suitable for customers who mainly want to purchase books and educational materials.

2. Target Customers and User Personars

2.1. Target Customers

The target customers of Clean Book include students, university students, and working professionals who want to purchase books online.

2.2. User Personars

This is Example Form:

Information	**Description**
Name	Nguyen Van A
Age	21
Occupation	University Student
Needs	Purchase textbooks and reference books for studying.
Difficulties	Has difficulty finding suitable books and comparing book information.
Goals	Find books quickly and place orders easily.
Behavior	Often searches for books online using a computer or smartphone.

- Nguyen Van A represents a university student who needs to find and purchase books conveniently. He prefers a website with clear categories, an easy-to-use search function, and a simple ordering process.

3. Feature Comparison for the Clean Book Project

Feature	Tiki	Fahasa
Product Search	Supports searching across various product categories.	Focuses on searching for books and related products.
Product Categories	Offers a wide range of product categories.	Organizes books and stationery into specialized categories.
Product Details	Provides information suitable for different product types.	Focuses on book information, such as title, author, and price.
Shopping Cart	Supports purchasing products from different categories.	Supports purchasing books and related products.
Shopping Interface	Displays various product categories and promotional content.	Focuses on books and stationery.

3.1. Advantages and Limitations of Tiki

Advantages:

- Offers a wide range of product categories.
- Serves customers with different shopping needs.
- Provides common e-commerce features.

Limitations:

- The large number of product categories may make navigation more complicated.
- A large amount of information and promotional content may distract some users.

3.2. Advantages and Limitations of Fahasa

Advantages:

- Focuses on books and stationery.
- Suitable for customers who mainly purchase books.
- Provides a more specialized shopping experience.

Limitations:

- Its product range is more specialized than that of a multi-category platform.
- It may not meet the needs of customers looking for products from many unrelated categories.

4. Selected Features for the Online Clean Book Project

Feature	Reason for Selection
User Registration and Login	Allows customers to create and access their accounts.
Browse Book Categories	Helps customers find books by category.

Search Books	Allows customers to find books quickly by title or keyword.
View Book Details	Provides important information before purchasing.
Shopping Cart	Allows customers to save selected books and manage quantities.
Place Order	Allows customers to complete their purchases online.
View Order History	Helps customers review their previous orders.
Manage Books	Allows administrators to add, update, and delete book information.
Manage Categories	Allows administrators to organize book categories.
Manage Orders	Allows administrators to view and update customer orders.

Note: Features may evolve as the project progresses to better meet user needs and project objectives.

5. UI/UX Observations and Reference Websites

5.1. Tiki

Tiki provides a multi-category shopping interface with various product sections and promotional content. Its search and category features help customers explore different products. However, the large amount of information may require users to spend more time finding specific products.

5.2. Fahasa

Fahasa provides an interface focused on books and stationery. Its specialized product categories help customers find books more easily. The website is suitable for customers who mainly purchase books and educational materials.

5.3. Lessons

Based on the observations of Tiki and Fahasa, Online Clean Book will focus on a simple and user-friendly interface. The website will provide clear book categories, an easy-to-use search function, detailed book information, and a straightforward shopping process.

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